

AMAN TAMBI

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science – Robotic Systems Development | (GPA: 3.9/4.0)

May 2027

Relevant Coursework: Robot Localization & Mapping; Manipulation, Estimation & Control; Robot Autonomy; Adv. Computer Vision; Robot Mobility

Birla Institute of Technology & Science, Pilani

Pilani, India

Bachelor of Engineering – Manufacturing Engineering | GPA: 8.69/10

May 2023

Relevant Coursework: Introduction to Robotics; Autonomous Mobile Navigation; Automation and Control

SKILLS

ML & Policy Learning: Deep Learning, Reinforcement Learning, Teacher Student Policy Distillation, VLMs/Foundation Models

Programming Languages: Python, C++, MATLAB, Embedded C

Frameworks: ROS2, PyTorch, Isaac Lab, MuJoCo, Gazebo, Fusion360, Deep Learning, OpenCV, Docker, Git, Linux

Hardware: RGB-D Cameras (Intel RealSense, ZED), Lidar (Ouster Os0), Franka Emika Panda, UR5, Hello Robot Stretch-3, Quadruped

PROJECTS

Lab Automation (*MRSD capstone project*)

Aug 2025 – May 2027

- Developing a mobile manipulator system for end-to-end well plate handling in wet-labs, integrating a motorized mobile base with a 6-DOF xArm for autonomous sample transfer between instruments (plate reader, liquid handler, incubator).
- Developing learning-based contact-rich manipulation for high-precision placing — training snap-fit insertion policies from force/torque and proprioceptive feedback as residual actions over admittance- and hybrid force-control priors, for robust placement under pose uncertainty.

Reactive Motion Planning with Generative Policies (*Independent research*)

Aug 2026 – Present

- Built a closed-loop rectified-flow motion planner for a 7-DoF Franka (point clouds in, 16-step joint-delta chunks out), trained on 12.2M trajectories. It beat a Neural MP baseline on all three official M π Nets test splits (79.8 % vs 48.0 % single-rollout success).
- Diagnosed its dominant failure: 80–90 % of collisions needed a ~20 cm detour the policy never made. Built a 440K-trajectory counterfactual dataset to teach it, as groundwork for RL fine-tuning toward dynamic obstacle avoidance without a hand-designed repulsion layer.

WORK EXPERIENCE

FieldAI

Irvine, USA

Robotics Research Intern-Locomotion & Planning

May 2026 – Aug 2026

- Developed perceptive locomotion for the Unitree A2 as a lower-cost alternative to the Boston Dynamics Spot (comparable specs at ~1/3 the cost), replacing the vendor's blind stock controller with in-house policies capable of terrain-aware traversal.
- Built a real-time 2.5D elevation mapping pipeline fusing onboard LiDAR with three depth cameras, producing 6 m × 6 m local height maps at 5 cm resolution and 10 Hz to supply exteroceptive input to downstream policies.
- Trained and deployed vision-based locomotion policies via teacher–student distillation with a GRU belief encoder that learns when to trust exteroceptive input versus fall back on proprioception, sustaining stable locomotion under noisy perception and sensor dropout; cut unintended limb contacts with stair risers and obstacles by 80% versus the blind baseline at matched traversal success.
- Characterized real-world elevation-map noise and parameterized it in simulation, injecting realistic map corruption during distillation to close the sim-to-real gap; deployed policies clear 40° slopes and 25 cm steps (≥ 0.2 m tread depth) on hardware.

Addverb

Noida, India

Robotics Engineer

Jan 2025 – Jun 2025

- Improved Model Predictive Control based walking control for quadrupeds and bipeds, enabling better balance recovery and robustness against external disturbances.
- Trained deep reinforcement learning policies in Isaac Lab using PyTorch, ensuring robust sim-to-real transfer and disturbance-tolerant quadruped locomotion up to 1.0 m/s on flat terrain.

Indian Institute of Technology, Delhi

Delhi, India

Project Scientist (Robotics Researcher)

May 2023 – Dec 2024

- Built an open set, zero-shot and dynamically updatable scene-graph generation pipeline using pre-trained visual language models with focused attention – F1=0.95(+0.16 v/s best baseline), stronger relation extraction and aided long horizon planning.
- Fused VLM segmentation masks and monocular-depth priors with multi-view point clouds for enhanced 3D modeling; demonstrated long horizon, collision-free sequential manipulation in unstructured and novel scenes
- Developed a ROS based autonomy stack combining perception, task planning, motion planning, open-world object grasping and error correction; Tested the stack on multiple mobile robots in both indoors and outdoors settings.